

TED UNIVERSITY

CMPE 491 (Senior Project I)

PROJECT SPECIFICATIONS REPORT

Project Name: ALPAR

(Agent Based Limit Aware Planning and Adaptive Routing)

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1. Introduction

ALPAR is an agent-based decision and planning architecture that replans safe routes for unmanned vehicles in real time when field conditions change and explains each routing decision to the operator in plain language. The project is carried out within the SAYZEK-ATP 2026–2027 program, which provides technical and academic mentorship and support for testing and validation.

Mission-critical unmanned systems, such as unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs), must operate safely in uncertain and dynamic environments. Classical route planners compute a route once, before the mission starts. They fall short when a new obstacle appears, a communication link is lost, a no-go zone is declared, or energy and budget limits tighten during the mission. In such cases the operator must either accept an unsafe route or replan by hand under time pressure. ALPAR addresses this gap with continuous monitoring, constraint-aware replanning and an explanation layer that makes each decision traceable.

1.1 Description

The aim of ALPAR is to deliver a domestically developed software solution that detects changes in the operating environment, replans a safe route within a bounded time, respects hard limits such as no-go zones and a minimum safe energy reserve, and reports why the selected route was preferred over its alternatives.

The proposal defines three contribution areas:

1. **Real-time replanning in dynamic crisis scenarios.** When risk exposure increases, a no-go zone is violated, a new obstacle is detected or communication is degraded, the system computes an updated route without restarting the mission.
2. **Explainable AI (XAI) decision support.** A language-model-based module produces a human-readable justification that compares the selected route with the main alternatives (for example, in terms of risk exposure, energy use, communication coverage and travel time). The goal is to increase operator trust and to make every decision auditable.
3. **Dynamic multi-criteria constraint management.** Route cost combines several criteria whose weights change with the mission state.

ALPAR is organized as a set of cooperating software agents around a shared mission and world model:

- **World and mission model** the map, obstacles, no-go zones, risk and threat areas, communication coverage, vehicle parameters (speed, energy consumption) and mission goals.
- **Monitoring agent** watches simulated sensor and status data and raises events such as a new obstacle, a new or expanded no-go zone, link loss or a drop in remaining energy.

- **Constraint and weighting agent** separates hard constraints (never violated) from soft criteria, and updates criterion weights according to the mission state (for example, energy weight rises as the reserve shrinks).
- **Planning agent** computes the route and a small set of feasible alternatives with established search-based planning algorithms, using the current multi-criteria cost.
- **Explanation (XAI) agent** turns the planner's computed metrics and the active constraints into a short, readable justification through a language model. Explanations are grounded in values the planner produced. The language model does not make the routing decision.
- **Operator interface** shows the map, the current and alternative routes, constraint status and the explanation, and lets the operator approve or reject a replanned route.

The architecture will be evaluated in a simulation environment against realistic crisis scenarios. Results will be compared with a static (plan-once) baseline in terms of constraint violations, replanning time, route quality and explanation quality. The project covers software design, implementation and simulation-based evaluation for UAV and UGV route planning.

1.2 Constraints

ALPAR is a software-only system evaluated in simulation, so its main constraints come from cost, safety, the defense-related application domain. Each constraint below is stated together with its effect on the design.

Economic Constraints

- The project has no dedicated hardware or licensing budget beyond the team's own computers and university resources.
- Commercial language-model APIs are billed per use, and repeated scenario runs can make this cost grow quickly. The design will keep the number of language-model calls low (one explanation per replanning event, not per planning step) and will consider locally hosted open-weight models.
- Computing power is limited to standard workstations. Planning algorithms and the simulation must run on such hardware without dedicated servers or high-end GPUs for core functionality.
- A domestically developed solution reduces long-term licensing and dependency costs for end users compared with imported route planning software.

Environmental Constraints

- Because ALPAR is software-only, its direct environmental impact is limited to the electricity used for development, simulation and language-model inference. Efficient algorithms and a small number of model calls keep this impact low.
- Simulation-based evaluation replaces many physical test flights or drives, which avoids their fuel or battery use, wear and risk.

- Treating energy as an explicit planning criterion leads to routes that use less energy in operation, which also reduces battery degradation over the vehicle's life.

Social Constraints

- Operators must be able to trust the system without trusting it blindly. Explanations must be short, clear and honest about uncertainty, so that they support the operator's judgment instead of encouraging automation bias.
- Explanations should be understandable by operators who are not experts in planning algorithms. They should use operational terms (risk, energy, link quality, time) rather than internal algorithm details.
- Public concern about autonomous systems in security applications is real. The project keeps a human operator in the loop and limits itself to route planning and decision support.

Political and Legal Constraints

- The application domain is defense-related and dual-use. The team will work only with open, synthetic or publicly available data, and will follow the confidentiality and data-handling rules of the SAYZEK-ATP program and the supervisor.
- The project supports national technology independence by producing a domestic software solution. Dependence on foreign cloud services for core functions should therefore be limited, and sensitive scenario data must not be sent to external services without approval.
- Real UAV operations in Türkiye are regulated by the Directorate General of Civil Aviation (SHGM). The simulated rules (no-go zones, altitude limits, flight restrictions) should be modeled so that they can later be mapped to such regulations, even though no real flight takes place in this project.
- Any personal data that may appear in the project (for example, user study participants evaluating explanations) must be handled in line with the Turkish Personal Data Protection Law (KVKK, Law No. 6698) [5].

Ethical Constraints

- The final decision on a replanned route stays with the human operator. The system recommends and explains; it does not remove human control.
- Explanations must be faithful to the planner's actual reasoning. The language model may only describe values and constraints computed by the planner, and must not invent reasons, numbers or alternatives.
- The project will not implement weapon release, targeting or any function intended to harm people.
- Limitations of the system (simulation-only validation, simplified models, possible failure cases) must be reported honestly in all project documents.

Health and Safety Constraints

- A wrong or late route in a real deployment could lead to a collision, a crash in a populated area or the loss of a vehicle. Safety-related limits (no-go zones, obstacle clearance, minimum energy reserve) are therefore modeled as hard constraints that the planner may never trade off against other criteria.
- When no route satisfies all hard constraints, the system must not return an unsafe route silently. It must alert the operator and propose a safe fallback, such as holding, returning to base or landing at the nearest safe point.
- Replanning must finish within a bounded time so that the recommendation is still valid when the operator sees it.
- All testing is done in simulation, so the project itself creates no physical risk to people.

Manufacturability Constraints

For a software product, manufacturability means that the system can be built, installed and reproduced reliably.

- The software must be modular, with clear interfaces between the world model, planner, constraint manager, explanation module and user interface, so that each part can be replaced or improved separately.
- The system must be installable on a standard workstation with documented, version-pinned dependencies, so that any team member, mentor or evaluator can reproduce the results.
- Interfaces should be designed so that the planner could later be connected to real vehicle autopilots or ground control software without rewriting the core.

Sustainability Constraints

- The code base must be documented, version-controlled and covered by automated tests so that it can be maintained after the project ends and continued within the SAYZEK program.
- Vehicle types, constraints and scenarios should be configurable rather than hard-coded, so that new platforms and mission types can be added without major redesign.
- The architecture should avoid lock-in to a single language-model provider, so that the explanation module keeps working if a provider changes its terms, prices or availability.

Time and Resource Constraints

- The project must be completed within two semesters (CMPE 491 and CMPE 492) by a team of three students. Requirements are therefore prioritized, and lower-priority features may be deferred.

- Access to real field data, real vehicles and real communication equipment is not available. Vehicle, energy and communication models will be simplified and their assumptions documented.

1.3 Professional and Ethical Issues

Because ALPAR recommends routes for vehicles operating near people and property, the team's central professional obligation is to put public safety first and to be honest about what the system can and cannot do. The team adopts the ACM Code of Ethics and Professional Conduct [1], the Software Engineering Code of Ethics and Professional Practice [2] and the IEEE Code of Ethics [3] as its guiding codes. Computer and information ethics literature [4] is used to reason about issues the codes do not settle directly.

Why this project needs explicit ethical attention? Moor's notion of a "policy vacuum", discussed in [4], describes situations where new computing capabilities appear before clear rules exist for their use. Agentic AI that replans routes and language models that justify decisions are such a case. The team therefore defines its own rules for the project in addition to following existing law and program rules.

Ethical Issues, Governing Principles and Project Measures

Table 1: Ethical issues and how ALPAR addresses them

Issue	Relevant principles	How ALPAR addresses it
Safety of people and property	ACM 1.2 Avoid harm [1]; SE Code PUBLIC [2]; IEEE I.1 (hold paramount the safety, health and welfare of the public) [3]	Safety limits are hard constraints; infeasible situations trigger an operator alert and a safe fallback; no physical tests in this project.
Human control over decisions	ACM 1.1, 1.2 [1]; IEEE I.2 (understanding of intelligent systems) [3]	The operator approves or rejects each replanned route; the system is decision support, not an autonomous decision maker.
Faithful and honest explanations	ACM 1.3 Be honest and trustworthy [1]; SE Code PRODUCT [2]; IEEE I.5 (honest and realistic claims) [3]	Explanations are generated only from planner outputs and active constraints; the team will check explanations for invented content and report the error rate.
Risk evaluation and transparency about limits	ACM 2.5 Give comprehensive and thorough evaluations, including risks [1]; IEEE I.1 (disclose factors that might endanger the public) [3]	Crisis scenarios, failure cases and modeling assumptions are documented; results are compared with a baseline, and limitations are stated in every report.
Quality of the software	ACM 2.1 Strive to achieve high quality [1]; SE Code PRODUCT [2]	Requirements traceability, code review, automated tests and simulation-based validation (Section 2).
Confidentiality and data handling	ACM 1.6 Respect privacy, 1.7 Honor confidentiality [1]; SE Code CLIENT AND EMPLOYER [2]	Only open or synthetic data; program and mentor confidentiality rules are followed; sensitive data is not sent to external AI services without approval.
Laws and rules	ACM 2.3 Know and respect existing rules [1]; IEEE I.4 (avoid unlawful conduct) [3]	KVKK for any personal data, aviation rules as modeling references, university academic integrity rules and SAYZEK program rules.

Issue	Relevant principles	How ALPAR addresses it
Intellectual property and licensing	ACM 1.5 Respect the work required to produce new ideas and artifacts [1]	Licenses of open-source libraries and models are checked and respected; all reused code, data and published methods are cited.
Competence	ACM 2.2 Maintain high standards of competence, 2.6 Perform work only in areas of competence [1]; SE Code SELF [2]; IEEE I.6 (maintain and improve technical competence) [3]	The team studies planning, multi-agent and XAI literature, and asks the supervisor and SAYZEK mentors for review in areas outside its experience.
Professional review	ACM 2.4 Accept and provide appropriate professional review [1]; SE Code JUDGMENT [2]	Design and results are reviewed at jury presentations and mentor meetings; critical feedback is recorded and answered.
Fair teamwork	SE Code COLLEAGUES and MANAGEMENT [2]; IEEE III (support colleagues in following the code) [3]	Tasks, responsibilities and credit are shared transparently among the three team members.

Professional Responsibilities of the Team

- **Responsibility to the public:** The team will not present ALPAR as ready for real operations. All results come from simulation with simplified models, and every report will state this clearly.
- **Responsibility to the supervisor, jury and SAYZEK:** The team will meet deadlines, report progress honestly, raise problems early and respect any confidentiality conditions of the program.
- **Honesty in reporting results:** Results will not be selected or adjusted to look better. Failed scenarios, constraint violations and limitations will be reported together with successes.
- **Responsible use of AI tools:** Any use of AI assistants in writing code or documents will follow university rules and will be disclosed where required. Team members remain responsible for checking and understanding everything they submit.
- **Dual-use awareness:** Route planning for unmanned systems has both civilian uses (search and rescue, disaster response, logistics) and security uses. The team limits its work to safe navigation and decision support and will raise concerns with the supervisor if the project is asked to move toward harmful functions.

2. Requirements

The initial requirements below define what ALPAR must do (functional), how well it must do it (non-functional) and the environment it runs in. Priorities follow the MoSCoW scheme: **Must** items are required for project success, **Should** items are planned but can be reduced if time runs short, and **Could** items are optional extensions. Numerical targets are initial values and will be refined after the first prototype and mentor feedback.

2.1 Functional Requirements

Mission and World Model

Table 2: Mission and world model functional requirements

ID	Requirement	Priority
FR-1	The system shall load a mission definition containing start point, goal point(s), vehicle type (UAV or UGV) and vehicle parameters (speed, energy capacity, energy consumption rate).	Must
FR-2	The system shall represent the operating area as a map with obstacles, no-go zones, risk or threat areas and communication coverage.	Must
FR-3	The system shall allow no-go zones, obstacles, risk areas and communication coverage to be added, removed or changed during a mission.	Must
FR-4	The system shall support at least one UAV model and one UGV model with different movement and energy characteristics.	Should

Monitoring and Event Detection

Table 3: Monitoring and event detection functional requirements

ID	Requirement	Priority
FR-5	The monitoring agent shall track vehicle position, remaining energy and communication link quality during the simulated mission.	Must
FR-6	The monitoring agent shall detect and report at least these events: new obstacle, new or expanded no-go zone, increased risk on the current route, communication loss or degradation, and energy reserve falling below a threshold.	Must
FR-7	Each detected event shall trigger a check of whether the current route is still valid and, if not, a replanning request.	Must

Planning and Replanning

Table 4: Planning and replanning functional requirements

ID	Requirement	Priority
FR-8	The planning agent shall compute an initial route from start to goal that satisfies all hard constraints.	Must
FR-9	The planning agent shall replan from the vehicle's current position when the current route becomes invalid or clearly worse than an alternative.	Must
FR-10	The planner shall treat no-go zones, obstacle clearance and the minimum safe energy reserve as hard constraints that are never violated.	Must
FR-11	The planner shall minimize a multi-criteria cost that includes at least risk exposure, energy consumption, communication quality and travel time or distance.	Must
FR-12	The planner shall produce, together with the selected route, at least two feasible alternative routes (when they exist) with their criterion values, for comparison and explanation.	Must
FR-13	If no feasible route exists, the system shall notify the operator and propose a safe fallback action (hold, return to base or land at the nearest safe point).	Must

ID	Requirement	Priority
FR-14	The system shall support at least two planning algorithms so that their performance can be compared in evaluation.	Could

Constraint and Weight Management

Table 5: Constraint and weight management functional requirements

ID	Requirement	Priority
FR-15	The constraint agent shall keep hard constraints separate from weighted soft criteria.	Must
FR-16	The constraint agent shall update criterion weights according to mission state, for example increasing the weight of energy as the reserve decreases and the weight of communication quality after a link degradation.	Must
FR-17	Weighting rules and thresholds shall be configurable without changing source code.	Should
FR-18	The operator shall be able to view the current weights and, within allowed limits, adjust mission priorities.	Should

Explanation (XAI) Module

Table 6: Explanation (XAI) module functional requirements

ID	Requirement	Priority
FR-19	For every replanning event, the system shall produce a natural-language explanation stating why replanning happened, which route was selected and why it was preferred over the alternatives.	Must
FR-20	Explanations shall be generated only from structured data produced by the planner and constraint agent (criterion values, active constraints, triggering event).	Must
FR-21	Every numerical value in an explanation shall match the corresponding value computed by the planner.	Must
FR-22	The system shall show a structured comparison (for example, a table of criteria for the selected and alternative routes) next to the text explanation.	Should
FR-23	If the language model is unavailable, the system shall still show the structured comparison and a template-based explanation.	Should

Operator Interface

Table 7: Operator interface functional requirements

ID	Requirement	Priority
FR-24	The interface shall display the map, vehicle position, current route, alternative routes, no-go zones, risk areas and communication coverage.	Must
FR-25	The interface shall show energy reserve, communication status and active constraints in real time.	Must
FR-26	The operator shall be able to approve or reject a replanned route; a rejected route shall not be executed.	Must
FR-27	The system shall keep a log of events, replanning decisions, explanations and operator actions for each mission.	Must

Simulation and Evaluation

Table 8: Simulation and evaluation functional requirements

ID	Requirement	Priority
FR-28	The simulation environment shall run predefined crisis scenarios, including at least: sudden no-go zone, new obstacle on route, increased risk exposure, communication loss and low energy.	Must
FR-29	The system shall run the same scenario with ALPAR and with a static (plan-once) baseline and record comparison metrics.	Must
FR-30	The evaluation tool shall report at least: number of hard constraint violations, mission success rate, replanning time, route length or time, energy used, time spent outside communication coverage and explanation accuracy.	Must
FR-31	Scenarios shall be reproducible from a configuration file and a random seed.	Must

2.2 Non-Functional Requirements

Table 9: Non-functional requirements

ID	Category	Requirement	Priority
NFR-1	Safety	Across all evaluation scenarios, routes approved by the operator shall contain zero hard constraint violations.	Must
NFR-2	Performance	Replanning (excluding explanation text generation) shall finish within 2 seconds on a standard workstation for the reference scenario size; the exact map size will be fixed in the design phase.	Must
NFR-3	Performance	The explanation shall be shown within 10 seconds of the replanning event.	Should
NFR-4	Reliability	Failure of one agent or of the language model shall not stop the simulation; the system shall report the failure and continue in a safe state.	Must
NFR-5	Explainability	In the evaluation set, explanations shall contain no invented reasons, routes or numerical values; any such error shall be counted and reported.	Must
NFR-6	Usability	An operator who has not used the system before shall be able to understand the route status and approve or reject a route after a short introduction; this will be checked with a small user evaluation.	Should
NFR-7	Maintainability	The system shall be modular, with documented interfaces between world model, agents, planner, explanation module and interface.	Must
NFR-8	Maintainability	Core planning and constraint functions shall be covered by automated unit tests.	Must
NFR-9	Portability	The system shall install and run on a standard workstation (Windows or Linux) using documented, version-pinned dependencies.	Must
NFR-10	Security	Scenario data shall not be sent to external services unless explicitly configured and approved; API keys shall not be stored in the code repository.	Must
NFR-11	Traceability	Every replanning decision shall be traceable from triggering event to selected route, explanation and operator action in the mission log.	Must
NFR-12	Independence	The explanation module shall work with at least one locally hosted language model so that core functions do not depend on a single external provider.	Should

3. References

[1] Association for Computing Machinery, "ACM Code of Ethics and Professional Conduct," 2018. [Online]. Available: https://www.acm.org/code-of-ethics

[2] ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices, "Software Engineering Code of Ethics and Professional Practice," Version 5.2, IEEE Computer Society and ACM, 1999. Also published in D. Gotterbarn, K. Miller, and S. Rogerson, "Software engineering code of ethics," *Communications of the ACM*, vol. 40, no. 11, pp. 110–118, 1997, doi: 10.1145/265684.265699. [Online]. Available: https://www.computer.org/education/code-of-ethics

[3] Institute of Electrical and Electronics Engineers, "IEEE Code of Ethics," IEEE Policies, Section 7.8. [Online]. Available: https://www.ieee.org/about/corporate/governance/p7-8.html

[4] T. W. Bynum, "Computer and Information Ethics," in *The Stanford Encyclopedia of Philosophy*, E. N. Zalta, Ed. Stanford, CA: Metaphysics Research Lab, Stanford University. [Online]. Available: https://plato.stanford.edu/entries/ethics-computer/

[5] Republic of Türkiye, Law on the Protection of Personal Data (KVKK), Law No. 6698, 2016.

[6] B. ÖZBEK, B. GİRAZ, and N. OYMAK, "ALPAR: Agent Based Limit Aware Planning and Adaptive Routing — Project Proposal," CMPE 491 Senior Project I, 2026. [Online]. Available: https://berenozbek.github.io/ALPAR_WebPage/